

DURBAN, South AfricaWMO: **685880**Lat: **30.013S**Lon: **30.926E**Elev: **46**StdP: **14.67**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.5	50.7	34.0	28.7	60.2	37.5	33.1	60.0	26.5	65.2	23.4	66.3	3.5	320	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.3	86.4	75.0	84.6	74.3	83.4	73.8	77.5	83.3	76.5	82.1	75.7	81.1	13.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.5	133.9	80.8	75.0	131.2	80.4	73.7	125.6	79.6	41.0	83.8	40.1	82.3	39.2	81.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.9	19.7	DB	43.7	93.5	2.5	3.1	41.9	95.7	40.4	97.5	39.0	99.2	37.1	101.4	(4)
(5)			WB	40.0	79.2	2.0	1.1	38.6	80.0	37.4	80.6	36.3	81.2	34.9	82.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.7	75.8	76.7	75.1	71.3	67.3	63.5	62.5	64.8	66.6	68.4	70.9	73.7	(6)	
	DBStd	5.87	3.43	3.27	3.36	3.60	3.36	3.04	3.04	3.36	3.87	3.94	3.88	3.71	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	258	0	0	0	2	16	63	89	44	27	13	4	0	(9)	
	CDD50	7189	800	748	779	640	536	406	388	459	497	572	628	736	(10)	
	CDD65	1969	335	328	314	192	87	18	11	37	75	120	181	271	(11)	
	CDH74	10352	2125	2348	1917	846	351	141	91	141	175	300	583	1334	(12)	
	CDH80	1594	360	452	314	97	31	20	11	17	15	34	62	181	(13)	
(14) Wind	WSAvg	8.9	9.7	9.3	8.8	8.4	6.9	6.7	7.2	8.9	9.9	10.4	10.3	9.9	(14)	
(15) Precipitation	PrecAvg	36.2	4.6	3.8	4.1	2.6	1.2	1.5	1.7	1.2	2.2	3.9	4.1	5.1	(15)	
	PrecMax	47.6	9.5	7.8	12.5	6.1	3.9	11.1	9.8	3.9	4.6	8.3	10.6	11.4	(16)	
	PrecMin	17.5	2.7	1.0	1.1	0.4	0.1	0.0	0.0	0.1	0.4	1.1	1.7	1.8	(17)	
	PrecStd	7.6	2.0	1.9	2.6	1.7	1.0	2.6	2.5	1.1	1.3	1.9	2.3	2.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.0	89.1	87.9	85.9	83.2	82.4	81.0	82.1	82.1	83.9	84.6	87.7	(19)	
		MCWB	76.3	77.0	75.6	73.0	66.1	61.8	61.6	64.0	68.9	71.0	73.1	75.9	(20)	
	2%	DB	85.7	86.2	84.8	82.5	79.8	77.3	76.0	77.3	78.4	79.4	81.8	84.1	(21)	
		MCWB	75.7	76.2	74.7	71.4	66.8	61.7	60.9	64.1	67.8	69.6	72.5	74.8	(22)	
	5%	DB	83.7	84.4	83.1	80.5	77.4	75.0	73.5	74.9	75.5	77.1	79.1	81.9	(23)	
		MCWB	74.8	75.3	74.0	70.9	66.7	62.2	61.3	64.1	66.5	68.7	71.2	73.5	(24)	
	10%	DB	82.1	82.7	81.5	78.7	75.5	73.0	71.7	72.8	73.6	75.1	77.1	79.9	(25)	
		MCWB	73.8	74.5	73.3	70.0	66.0	61.5	61.1	63.5	65.6	67.6	70.1	72.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.6	79.0	78.0	75.1	71.5	67.6	66.4	68.6	70.9	72.6	75.4	77.7	(27)	
		MCDB	84.8	85.2	83.5	80.7	77.3	74.2	71.9	74.7	77.9	79.8	81.2	84.2	(28)	
	2%	WB	77.1	77.8	76.4	73.4	69.9	65.8	64.9	67.2	69.2	70.9	73.9	76.2	(29)	
		MCDB	82.8	83.6	82.1	79.4	75.6	72.5	71.5	73.1	75.3	77.0	79.2	81.7	(30)	
	5%	WB	76.0	76.7	75.5	72.1	68.7	64.5	63.7	66.1	68.1	69.7	72.6	74.7	(31)	
		MCDB	81.5	82.1	81.1	78.0	74.8	71.6	70.8	72.1	73.4	75.0	77.5	79.8	(32)	
	10%	WB	74.9	75.9	74.4	71.1	67.4	63.0	62.4	65.0	67.0	68.8	71.3	73.5	(33)	
		MCDB	80.3	81.1	80.0	76.9	73.8	70.2	69.6	71.1	72.4	73.8	76.0	78.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	11.3	11.8	13.7	17.1	20.1	19.1	16.1	13.3	11.4	10.9	10.7	(35)	
		MCDBR	13.0	13.2	13.8	16.4	20.1	24.3	23.6	20.7	16.5	14.7	13.8	13.2	(36)	
	5% WB	MCWBR	6.8	6.7	7.3	8.6	11.6	14.3	14.5	12.1	9.6	8.3	8.0	7.4	(37)	
		MCWBR	12.1	12.1	12.7	14.7	17.5	20.5	20.7	17.7	14.6	12.9	12.6	12.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.456	0.447	0.438	0.407	0.381	0.365	0.375	0.438	0.487	0.496	0.470	0.463	(40)	
		taud	2.301	2.333	2.339	2.389	2.424	2.446	2.406	2.218	2.099	2.116	2.228	2.267	(41)	
	Ebn at Noon		282	279	270	262	253	251	252	247	251	262	276	281	(42)	
		Edh at Noon	44	42	39	34	30	27	30	39	49	51	48	46	(43)	
(44) All-Sky Solar Radiation	RadAvg		1876	1834	1594	1297	1105	984	1050	1225	1413	1549	1709	1840	(44)	
	RadStd		150	136	77	83	57	48	70	75	90	142	138	135	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page